Assembly and Characterization of a Low-Resolution Prism Spectrograph (BESOS): Range, Dispersion and Resolution

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I. Introduction

A spectrograph disperses the light of a source in wavelength, giving access to information that an image does not carry: chemical composition, temperature, radial velocity, or whether the emission is a continuum or a set of lines. The basic layout is a slit, a collimator, a dispersing element and a camera that focuses each wavelength at a different position of the detector.

The instrument assembled in this experiment is BESOS (*BEst Simple Optical Spectrograph*), a low-resolution design by the CAOS group [2]. Its dispersing element is an equilateral SF18 prism, so the dispersion comes from the variation of the refractive index with wavelength. Compared with a grating, a prism gives lower resolution and a strongly non-linear dispersion (larger in the blue than in the red), but it sends all the light into a single spectrum.

The goal is to assemble the instrument and to measure in the laboratory its **spectral range**, **dispersion** and **spectral resolution**, comparing them with the first-order expectation [1].

II. Theoretical Framework

A. Prism equation

A ray entering a prism of apex angle A and index $n(\lambda)$ at incidence θ_1 is refracted at both faces ($\sin\theta_1 = n\sin r_1$, $r_1 + r_2 = A$, $\sin\theta_2 = n\sin r_2$), and is deviated by [3]

$$\delta(\lambda) = \theta_1 - A + \arcsin\Big[n\sin\Big(A - \arcsin\frac{\sin\theta_1}{n}\Big)\Big]. \quad (1)$$

The prism is mounted near minimum deviation, where the beam crosses it symmetrically ($\theta_1 = \theta_2$) and $n = \sin[(A + \delta_{\min})/2]/\sin(A/2)$. The index follows the Sellmeier law $n^2 = 1 + \sum_i B_i\lambda^2/(\lambda^2 - C_i)$.

B. Dispersion

Differentiating (1) at fixed θ_1,

$$\frac{\mathrm{d}\delta}{\mathrm{d}\lambda} = \frac{\sin A}{\cos r_1\cos\theta_2}\frac{\mathrm{d}n}{\mathrm{d}\lambda} \overset{\delta_{\min}}{=} \frac{2\sin(A/2)}{\sqrt{1 - n^2\sin^2(A/2)}}\frac{\mathrm{d}n}{\mathrm{d}\lambda}. \quad (2)$$

The camera of focal length f_{cam} turns angles into positions, so the reciprocal dispersion on a detector of pixel size p is

$$P \equiv \left|\frac{\mathrm{d}\lambda}{\mathrm{d}x}\right| = \frac{p}{f_{cam}\,|\mathrm{d}\delta/\mathrm{d}\lambda|} \quad [\mathrm{nm/px}]. \quad (3)$$

Experimentally, $\lambda(x)$ of a prism is well described by the Hartmann formula [4],

$$\lambda(x) = \lambda_0 + \frac{C}{x - x_0} \;\Rightarrow\; P(x) = \frac{|C|}{(x - x_0)^2}, \quad (4)$$

with λ_0, C and x_0 fitted to lines of known wavelength.

C. Resolution

A slit of width w is imaged on the detector with width $w' = w\,(f_{cam}/f_{col})\,r$, where $r = \cos\theta_1\cos r_2/(\cos r_1\cos\theta_2)$ is the anamorphic magnification of the prism ($r = 1$ at minimum deviation). The resolution element and the resolving power are

$$\Delta\lambda = \max(w', 2p)\,\frac{P}{p}, \qquad R = \frac{\lambda}{\Delta\lambda} \overset{w'\geq 2p}{=} \frac{\lambda f_{col}}{w\,r}\left|\frac{\mathrm{d}\delta}{\mathrm{d}\lambda}\right|, \quad (5)$$

so a slit-limited R does not depend on f_{cam}. In the measurement, $\Delta\lambda = \mathrm{FWHM}_{px}\,P(\lambda)$ for a narrow line. The diffraction limit of the prism is $R_{dif} = b\,|\mathrm{d}n/\mathrm{d}\lambda|$, with b the illuminated base.

D. Spectral range

With λ_c at the centre of a detector of width $L = N_x p$, the covered range is set by

$$\big|f_{cam}\tan[\delta(\lambda) - \delta(\lambda_c)]\big| \leq L/2, \quad (6)$$

and is further limited by the transmission of the optics, the quantum efficiency of the detector and the spectrum of the source.

E. Expected performance

Equations (1)–(6) were evaluated with the design values: $A = 60°$, SF18 with the SCHOTT Sellmeier coefficients [6] ($n = 1.7267$ at 550 nm), $f_{col} = 30\,\mathrm{mm}$, $f_{cam} = 40\,\mathrm{mm}$, $w = 25\,\mu\mathrm{m}$, $p = 4.65\,\mu\mathrm{m}$ and minimum deviation at 550 nm ($\theta_1 = 59.7°$, $\delta_{\min} = 59.4°$). The slit image is $w' = 33.3\,\mu\mathrm{m}$ (7.2 px), P goes from 0.26 nm/px at 500 nm to 0.52 nm/px at 600 nm, and R falls from 197 at 557 nm to 77 at 878 nm, well below $R_{dif} \approx 3241$, so the design is slit limited. With 550 nm centred on the 5.95 mm detector, (6) gives 416 nm to beyond 1000 nm. These values are compared with the measurements in Table IV.

III. Equipment and Method

Table I lists the equipment. BESOS was assembled following the CAOS drawings [2], [1] (Fig. 1), with the 30 mm doublet as collimator and the 40 mm doublet as camera objective. A 50 mm lens placed about 50 mm in front of the slit concentrated the light of each source on it. The camera was focused with the red LED placed far from the instrument, minimizing the width of its slit image.

TABLE I: Equipment used in the experiment.

Item	Use
BESOS head: slit 25 µm×3 mm; collimator doublet $f = 30$ mm; SF18 prism, $A = 60°$, 15 mm; camera doublet $f = 40$ mm	instrument under test
Lens $f = 50$ mm	feeds the slit
DMK 41AU02.AS (Sony ICX205AL CCD, 1280 × 960, 4.65 µm, 8 bit)	spectrum camera
DMK 21AU04.AS (640 × 480, 5.6 µm)	slit viewer (alignment)
Blue, green and red LEDs	alignment, calibration check
Ar and Kr spectral lamps	wavelength calibration, resolution
Phone LED flash	continuum source (range)
AstroDMx Capture 2.17.2 [7]	FITS acquisition
Python (numpy, scipy, astropy)	data reduction

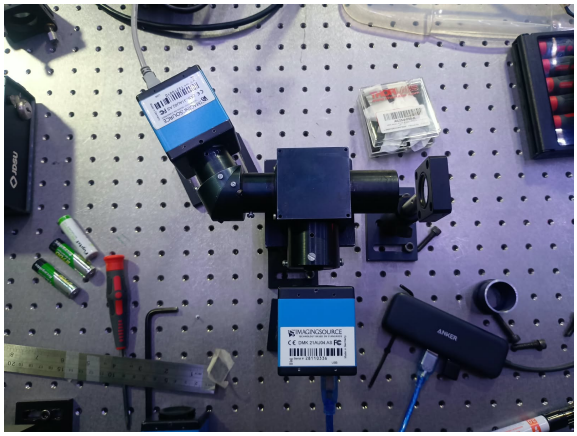


Fig. 1: Setup: BESOS head (centre), spectrum camera DMK 41AU02.AS on the inclined port (top left), slit-viewer camera DMK 21AU04.AS (bottom) and the feeding lens mount (right).

For each source 20 frames were taken in MONO8 format (gain 354–392, gamma 104, 1 ms to 150 ms exposure, chosen to avoid saturation except for the flash). No dark frames were taken; the background was estimated on the unilluminated rows of each frame. The reduction, implemented in a Python package, proceeds as follows: (i) median combination of the frames; (ii) tracing of a Kr line row by row and quadratic fit $x(y)$, used to shift every row so that the curved lines become vertical; (iii) extraction of the 1D spectrum averaging rows 450–550, the best-focus region; (iv) Gaussian fits, simultaneous for blended features; (v) Hartmann calibration (4) with isolated Ar and Kr lines (air wavelengths from NIST [5]); (vi) computation of P, R and the range, and comparison with (1)–(6).

IV. Results

A. Line curvature and extraction

Dispersion runs along x, blue to the left. The slit images are not straight but parabolic, with a sagitta of ∼42 px between rows 250 and 800 and a mean tilt of $-2.2°$ in the extracted region. The quadratic trace fits the centroids with 0.92 px rms and removes the curvature (Fig. 2). The line width also changes along the slit (Fig. 3): it is minimum, ∼3 px, in rows 450–550 and grows to 5–7 px towards the ends, so only that band was extracted.

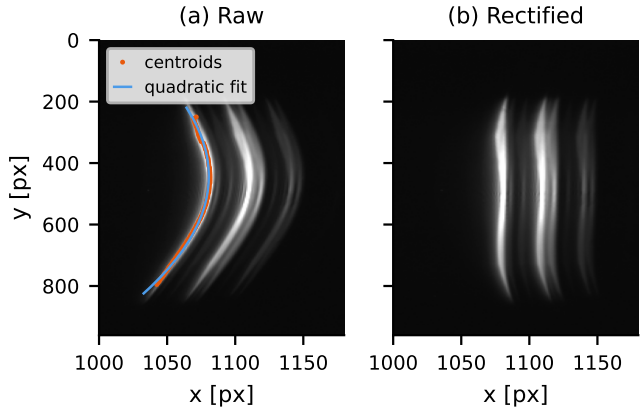


Fig. 2: Kr lines around 760–830 nm. (a) Raw frame with the centroids of the traced line and the quadratic fit. (b) Same region after rectification.

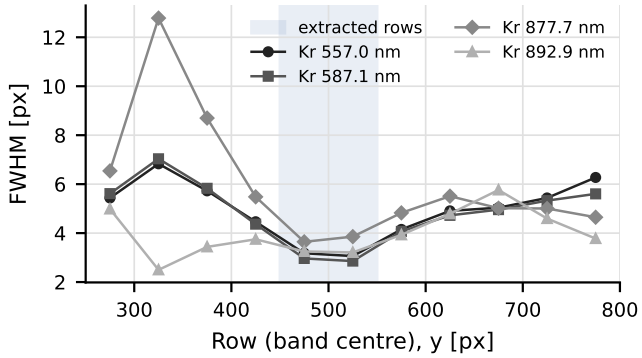


Fig. 3: FWHM of four isolated Kr lines measured in 50-row bands along the slit. The shaded band marks the extracted rows.

B. Line identification and wavelength calibration

No specifications were available for the lamps or the LEDs. The lamps were identified from their line pattern: the white lamp is krypton (weak lines at 557.0 and 587.1 nm, strong ones at 758–830 nm) and the yellowish one is argon (696.5, 706.7,

763.5, 811.5, 912.3, 965.8 nm). Fig. 4 shows the profiles and Gaussian fits of all sources. The LEDs are 20–27 nm wide, far broader than the atomic lines, so they are used only as an independent check. Of the 14 fitted lamp features, 7 are isolated lines; the rest are unresolved blends and were not used.

The Hartmann fit to the 7 isolated lines gives $\lambda_0 = 246.93$ nm, $C = -162304$ nm px and $x_0 = 1396.9$ px, with residuals of 0.35 nm rms (Fig. 5, Table II). Applied to the LEDs, which were not in the fit, it places them at 464.2, 523.6 and 629.2 nm (Table III), consistent with typical blue, green and red LEDs. This also checks the blue end, where the calibration is an extrapolation.

TABLE II: Isolated lines: laboratory wavelength [nm], centroid, calibration residual [nm], FWHM, reciprocal dispersion, resolution element and resolving power.

Lamp	$\lambda_{\rm lab}$	x [px]	Resid.	FWHM [px]	P [nm/px]	$\Delta\lambda$ [nm]	R
Kr	557.0	873.7	−0.07	3.1	0.59	1.9±0.1	301
Kr	587.1	920.0	−0.13	2.9	0.71	2.1±0.1	283
Ar	696.5	1035.7	+0.34	3.7	1.24	4.6±0.3	153
Ar	706.7	1043.8	+0.15	2.8	1.30	3.7±0.8	191
Kr	877.7	1139.7	−0.35	3.8	2.45	9.2±0.4	95
Kr	892.9	1145.8	−0.29	3.2	2.57	8.3±0.7	107
Ar	965.8	1171.0	+0.36	5.3	3.18	16.8±0.8	58

TABLE III: LED positions from the calibration (fit error only).

Source	$\lambda_{\rm nom}$	$\lambda_{\rm meas}$ [nm]	FWHM [nm]
Blue LED	465	464.2±0.6	19.4
Green LED	525	523.6±0.4	27.4
Red LED	625	629.2±0.2	21.1

C. Dispersion

The measured reciprocal dispersion is the derivative of the calibration, $P = |C|/(x - x_0)^2$ (Fig. 6). It grows from 0.39 nm/px at 500 nm to 0.77 nm/px at 600 nm and 1.88 nm/px at 800 nm. Its shape follows the SF18 prism (the ratio to the model varies only 3% over 440–970 nm), but its value is 1.47 times the design one: the spectrum is more compressed than expected, as if f_{cam} were 27.3 mm instead of 40 mm.

D. Resolution

The isolated Kr lines have a median FWHM of 3.2 px and the Ar lines 3.7 px. The resolution element grows from about 2 nm at 557–587 nm to 17 nm at 966 nm, and the resolving power falls from $R \approx 301$ to $R \approx 58$ (Table II, Fig. 7). The measured values are above the design curve at all wavelengths covered by Kr.

E. Spectral range

The flash spectrum (Fig. 8) shows the pump LED peak near 450 nm, the dip near 480 nm and the broad phosphor band. With a 5% threshold the range is 424–726 nm. 6.5% of the flash pixels are saturated, which flattens 520–610 nm but does not move the edges. The calibration places the blue edge of the detector ($x = 0$) at 363 nm, and the Ar line at 966 nm was detected at the red end, so the instrument covers at least 363–966 nm and the flash range is set by the source.

TABLE IV: Summary: first-order design vs. measured values.

Parameter	Design	Measured
P at 500 nm [nm/px]	0.26	0.39
P at 600 nm [nm/px]	0.52	0.77
P at 800 nm [nm/px]	1.30	1.88
Slit image FWHM [px]	7.2	3.2
$\Delta\lambda$ at 557 nm [nm]	2.8	1.9
R at 557 nm	197	301
R at 878 nm	77	95
Range [nm]	416–>1000[a]	424–726[b]

[a] Detector limit with 550 nm centred. [b] Flash, limited by the source; detector covers at least 363–966 nm.

V. Discussion

Dispersion and line width. The reciprocal dispersion was found to be 1.47 times the design value, and the lines (3.2 px) are narrower than the expected slit image (7.2 px). Both effects can be explained by a prism rotated away from minimum deviation: at $\theta_1 \approx 74^\circ$ the anamorphic factor $r = 0.44$ reproduces the measured line width and accounts for a factor 1.21 in P. The remaining ∼20% is attributed to the real focal length of the camera doublet and the detector position, which were not measured.

Resolution. R lies above the design curve because the narrower lines more than compensate the lower dispersion. The instrument is slit limited (R_{dif} is ten times larger) and the ∼3 px FWHM satisfies Nyquist sampling. The Ar values are the least reliable, due to blends and a less uniform illumination of the slit.

Curvature. The parabolic lines are expected in a prism spectrograph, since rays from the ends of the slit cross the prism out of the principal plane and are deviated more. Rectification removes this effect, and the change of FWHM along the slit (Fig. 3) indicates field curvature or detector tilt, so the quoted R refers to the centre of the slit.

Range. The flash range is limited by the source; the instrument covers at least 363–966 nm. In the near infrared $P > 2$ nm/px (700–970 nm span only 134 px), so the useful range is the visible.

Error sources. The blue calibration is an extrapolation (no lamp lines below 557 nm) checked only with LEDs of unknown wavelength, and the lamps were identified by their line pattern. The 8-bit format, the missing darks and the saturated flash affect the profiles more than the centroids.

VI. Conclusions

The BESOS prism spectrograph was assembled and characterized with LEDs, Kr and Ar lamps and a phone flash. A Hartmann calibration on 7 isolated lines reaches 0.35 nm rms and is confirmed independently by the three LEDs. The reciprocal dispersion is strongly non-linear, from 0.39 nm/px at 500 nm to 1.88 nm/px at 800 nm. It has the shape expected for SF18 but is 1.47 times the design value. The resolving power is slit limited, from $R \approx 301$ at 557 nm to $R \approx 100$ at

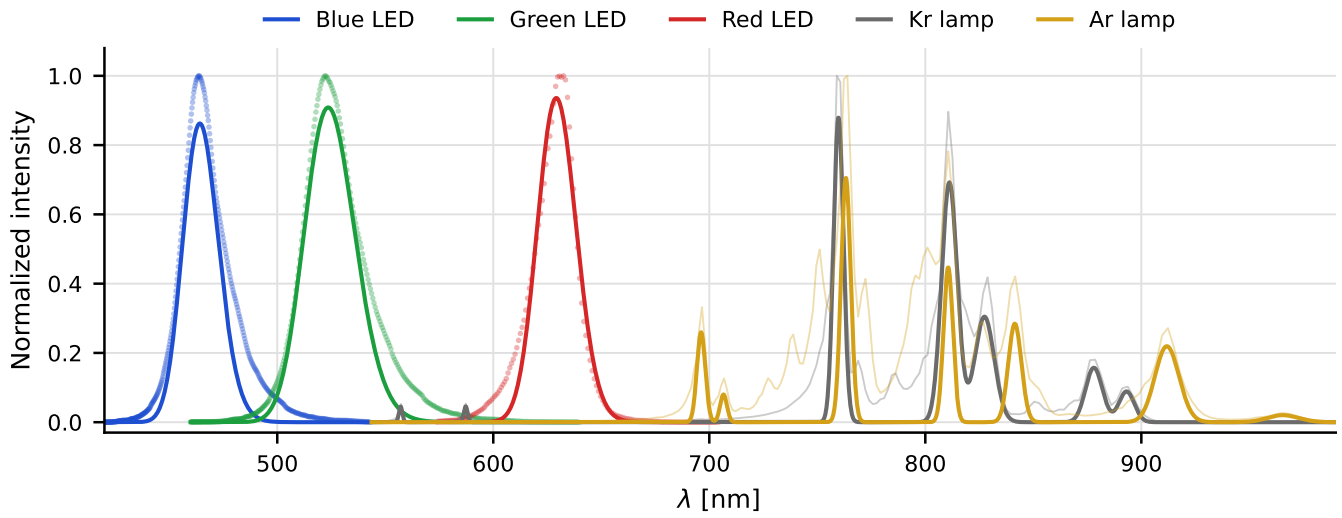


Fig. 4: Spectral profiles (faint) and Gaussian fits (solid) of the three LEDs and of the Kr and Ar lamps, normalized to the peak of each source and plotted with the final calibration.

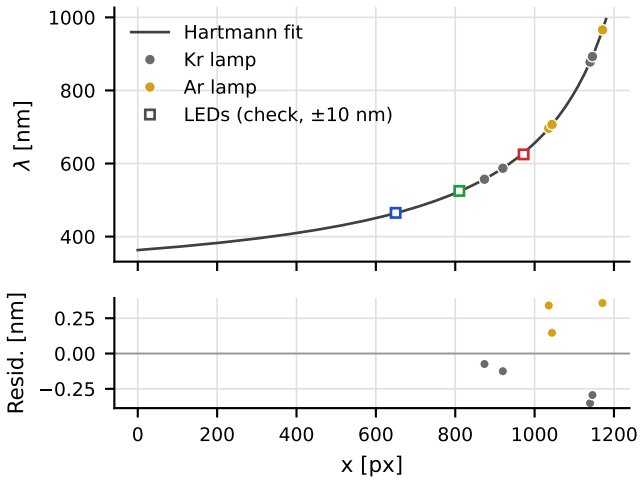


Fig. 5: Hartmann calibration with the isolated Kr and Ar lines (top) and its residuals (bottom). The LEDs (squares, nominal $\lambda \pm 10$ nm) are not part of the fit.

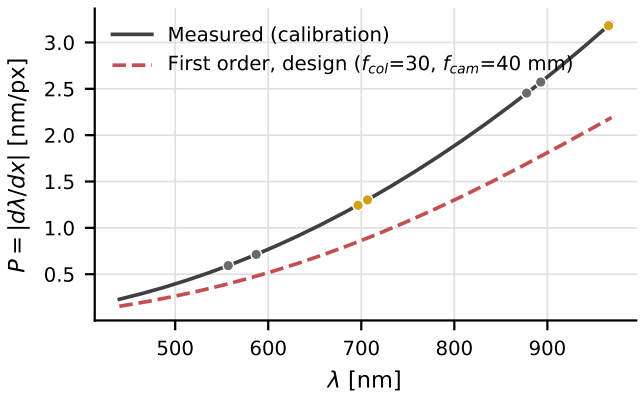


Fig. 6: Measured reciprocal dispersion (derivative of the calibration; dots at the calibration lines) and first-order design value.

880–890 nm, above the design expectation because the lines are narrower than the nominal slit image. The range measured with the flash is 424–726 nm, limited by the source, while the instrument covers at least 363–966 nm.

The objective of the experience was met: range, dispersion and resolution were measured and compared with theory. The two departures, compressed dispersion and narrow lines, are both consistent with a prism away from minimum deviation plus an uncertain camera scale. Measuring the deviation angle, the real slit width and the doublet focal lengths, and adding a lamp with blue lines are the next steps.

References

[1] L. Vanzi, *Astronomical Instrumentation Lab IEE3873 – Construction of a small low resolution spectrometer for the visible*, course guidelines and *Introduction to Low Resolution Spectrograph*, AIUC, Pontificia Universidad Católica de Chile.

[2] CAOS – Club of Amateurs in Optical Spectroscopy, *BESOS: a prism spectrograph*. https://spectroscopy.wordpress.com/2011/04/10/besos-a-prism-spectrograph/

[3] J. F. James, *Spectrograph Design Fundamentals*. Cambridge University Press, 2007.

[4] J. Hartmann, "A simple interpolation formula for the prismatic spectrum," *Astrophys. J.*, vol. 8, p. 218, 1898.

[5] A. Kramida, Yu. Ralchenko, J. Reader and NIST ASD Team, *NIST Atomic Spectra Database*. https://physics.nist.gov/asd

[6] SCHOTT AG, optical glass data sheet, SF18 ($n_d = 1.72151$, $V_d = 29.25$).

[7] N. Palmer and S. Wainwright, *AstroDMx Capture*, v2.17.2. https://www.astrodmx-capture.org.uk

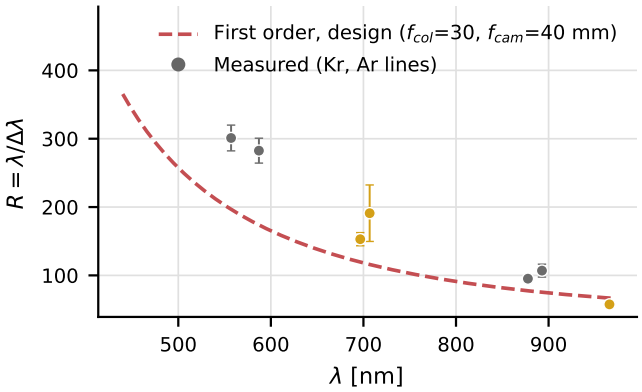


Fig. 7: Measured resolving power (Kr grey, Ar yellow) and first-order design value for a 25 µm slit.

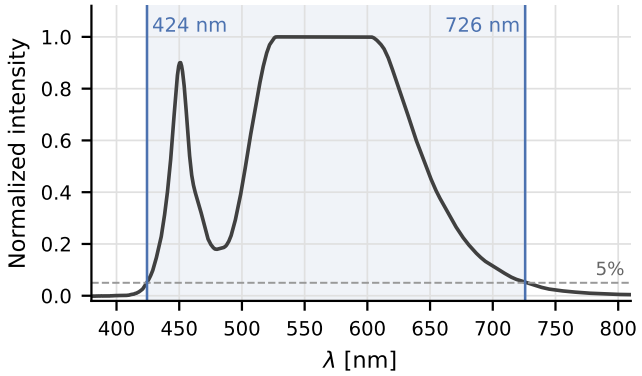


Fig. 8: Phone flash spectrum and range defined at 5% of the peak.